



МИНОБРНАУКИ РОССИИ

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ОТЧЕТ ПО ПРАКТИЧЕСКОЙ РАБОТЕ №3
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1 ЗАПУСК И БАЗОВАЯ НАСТРОЙКА FLUME

1.1 Простой перенос данных через порт

Процесс создания директории для flume представлен на Рисунке 1.

```
[root@osboxes ~]# mkdir ~/flume
[root@osboxes ~]# cd ~/flume
[root@osboxes flume]# ls
[root@osboxes flume]# vi transfer.conf
```

Рисунок 1 — Директория для flume

Настройка файла конфигурации agent1 представлена на Рисунке 2.

```
[root@osboxes flume]# cat transfer.conf

agent1.sources = netcatSrc
agent1.channels = memChannel
agent1.sinks = log
agent1.sources.netcatSrc.channels = memChannel
agent1.sinks.log.channel = memChannel
agent1.sources.netcatSrc.type = netcat
agent1.sources.netcatSrc.bind = 0.0.0.0
agent1.sources.netcatSrc.port = 3333
agent1.sinks.log.type = logger
agent1.channels.memChannel.type = memory
agent1.channels.memChannel.capacity = 100
...

[root@osboxes flume]# |
```

Рисунок 2 — Конфигурация agent1

Процесс запуска агента представлен на Рисунке 3.

```
[root@osboxes flume]# flume-ng agent -name agent1 --conf-file transfer.conf
Warning: No configuration directory set! Use --conf <dir> to override.
Info: Including Hadoop libraries found via (/home/hadoop/hadoop/bin/hadoop) for HDFS access
Info: Including HBASE libraries found via (/usr/local/hbase/hbase-2.3.5/bin/hbase) for HBASE access
Info: Including Hive libraries found via O for Hive access
+ exec /usr/lib/jvm/java-1.8.0-openjdk/bin/java -Xmx20m -cp "/usr/local/flume/flume-1.9.0/lib/*:/home/hadoop/hadoop/etc/hadoop:/home/hadoop/hadoop/share/hadoop/common/lib/*:/home/hadoop/hadoop/share/hadoop/common/*:/home/hadoop/hadoop/share/hadoop/hdfs:/home/hadoop/hadoop/share/hadoop/hdfs/lib/*:/home/hadoop/hadoop/share/hadoop/hdfs/*:/home/hadoop/hadoop/share/hadoop/mapred/*:/home/hadoop/hadoop/share/hadoop/yarn:/home/hadoop/hadoop/share/hadoop/yarn/lib/*:/home/hadoop/hadoop/share/hadoop/yarn/*:/usr/local/hbase/hbase-2.3.5/conf:/usr/lib/jvm/jre/lib/tools.jar:/usr/local/hbase/hbase-2.3.5:/usr/local/hbase/hbase-2.3.5/lib/shaded-clients/hbase-shaded-client-byo-hadoop-2.3.5.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/audience-annotations-0.5.0.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/commons-logging-1.2.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/htrace-core4-4.2.0-incubating.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/log4j-1.2.17.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/slf4j-api-1.7.30.jar:/home/hadoop/hadoop/etc/hadoop:/home/hadoop/hadoop/share/hadoop/common/lib/*:/home/hadoop/hadoop/share/hadoop/common/*:/home/hadoop/hadoop/share/hadoop/hdfs:/home/hadoop/hadoop/share/hadoop/hdfs/lib/*:/home/hadoop/hadoop/share/hadoop/hdfs/*:/home/hadoop/hadoop/share/hadoop/mapreduce/*:/home/hadoop/hadoop/share/hadoop/yarn:/home/hadoop/hadoop/share/hadoop/yarn/lib/*:/home/hadoop/hadoop/share/hadoop/yarn/*:/usr/local/hbase/hbase-2.3.5/conf:/lib/*" -Djava.library.path="/home/hadoop/hadoop/lib/native:/home/hadoop/hadoop/lib/org.apache.flume.node.Application -name agent1 --conf-file transfer.conf
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/flume/flume-1.9.0/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hadoop/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.30.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2025-10-08 04:13:53,219 INFO node.PollingPropertiesFileConfigurationProvider: Configuration provider starting
2025-10-08 04:13:53,225 INFO node.PollingPropertiesFileConfigurationProvider: Reloading configuration File:transfer.conf
2025-10-08 04:13:53,229 INFO conf.FlumeConfiguration: Processing:log
2025-10-08 04:13:53,230 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 04:13:53,232 WARN conf.FlumeConfiguration: Configuration property ignored: '...' =
2025-10-08 04:13:53,232 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 04:13:53,232 INFO conf.FlumeConfiguration: Processing:memChannel
2025-10-08 04:13:53,232 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 04:13:53,232 INFO conf.FlumeConfiguration: Processing:log
2025-10-08 04:13:53,233 INFO conf.FlumeConfiguration: Added sinks: log Agent: agent1
2025-10-08 04:13:53,233 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 04:13:53,233 WARN conf.FlumeConfiguration: Agent configuration for 'agent1' has no configfilters.
2025-10-08 04:13:53,265 INFO conf.FlumeConfiguration: Post-validation flume configuration contains configuration for agents: [agent1]
2025-10-08 04:13:53,265 INFO node.AbstractConfigurationProvider: Creating channels
2025-10-08 04:13:53,275 INFO channel.DefaultChannelFactory: Creating instance of channel memChannel type memory
2025-10-08 04:13:53,281 INFO node.AbstractConfigurationProvider: Created channel memChannel
2025-10-08 04:13:53,281 INFO source.DefaultSourceFactory: Creating instance of source netcatSrc, type netcat
2025-10-08 04:13:53,288 INFO sink.DefaultSinkFactory: Creating instance of sink: log, type: logger
2025-10-08 04:13:53,292 INFO node.AbstractConfigurationProvider: channel memChannel connected to [netcatSrc, log]
2025-10-08 04:13:53,300 INFO node.Application: Starting new configuration:{ sourceRunners:{netcatSrc=EventDrivenSourceRunner: { source:org.apache.flume.source.NetcatSource{name:netcatSrc,sinks:org.apache.flume.sink.DefaultSinkProcessor@80c526 counterGroup:{ name:null counters:{} } } } channels:{memChannel=org.apache.flume.channel.MemoryChannel{name:memChannel}}}
2025-10-08 04:13:53,305 INFO node.Application: Starting Channel memChannel
2025-10-08 04:13:53,597 INFO instrumentation.MonitoredCounterGroup: Monitored counter group for type: CHANNEL, name: memChannel: Successfully registered new MBean.
2025-10-08 04:13:53,597 INFO instrumentation.MonitoredCounterGroup: Component type: CHANNEL, name: memChannel started
2025-10-08 04:13:53,598 INFO node.Application: Starting Sink log
2025-10-08 04:13:53,601 INFO node.Application: Starting Source netcatSrc
2025-10-08 04:13:53,605 INFO source.NetcatSource: Source starting
2025-10-08 04:13:53,622 INFO source.NetcatSource: Created serverSocket:sun.nio.ch.ServerSocketChannelImpl[/0:0:0:0:0:0:0:0:3333]
```

Рисунок 3 —Запуск агента

Установка и отправка данных через telnet представлена на Рисунке 4.

```
[root@osboxes ~]# yum install telnet
Loaded plugins: langpacks
base                                     3.6 kB 00:00:00
epel/x86_64/metalink                   4.4 kB 00:00:00
extras                                  2.9 kB 00:00:00
mariadb                                 3.4 kB 00:00:00
updates                                 2.9 kB 00:00:00
Resolving Dependencies
--> Running transaction check
---> Package telnet.x86_64 1:0.17-66.e17 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package                               Arch              Version              Repository              Size
=====
Installing:
telnet                                x86_64            1:0.17-66.e17        updates                  64 k
Transaction Summary
=====
Install 1 Package
Total download size: 64 k
Installed size: 111 k
Is this ok [y/d/N]: y
Downloading packages:
telnet-0.17-66.e17.x86_64.rpm          | 64 kB 00:00:00
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Installing : 1:telnet-0.17-66.e17.x86_64
  Verifying  : 1:telnet-0.17-66.e17.x86_64
Installed:
telnet.x86_64 1:0.17-66.e17

Complete!
[root@osboxes ~]# telnet localhost 3333
Trying ::1...
Connected to localhost.
Escape character is '^['.
Skillnet
OK
Hadoop
OK
OK
ACConnection closed by foreign host.
```

Рисунок 4 — Отправка данных через telnet

Проверка отображения данных в агенте представлена на Рисунке 5.

```
2025-10-08 04:16:55,726 INFO sink.LoggerSink: Event: { headers:{} body: 53 68 69 6C 6C 65 74 0D skillnet. }
2025-10-08 04:16:55,726 INFO sink.LoggerSink: Event: { headers:{} body: 48 61 64 6F 6F 70 0D Hadoop. }
2025-10-08 04:16:55,726 INFO sink.LoggerSink: Event: { headers:{} body: 0D . }
```

Рисунок 5 — Проверка отправки данных

Завершение работы агента представлено на Рисунке 6.

```

Ac2025-10-08 04:19:48,594 INFO node.Application: Shutting down configuration: { sourceRunners:{netcatSrc=EventDrivenSourceRunner: { source:org.apache.flume.source.NetcatSource{name:netcatSrc,state:START} }} sinkRunners:{log=SinkRunner: { policy:org.apache.flume.sink.DefaultSinkProcessor@7801c526 counterGroup:{ name:null counters:{runner.backoffs.consecutive=23, runner.backoffs=47} }} } } channels:{memChannel=org.apache.flume.channel.MemoryChannel{name: memChannel}} }
2025-10-08 04:19:48,595 INFO node.Application: Stopping Source netcatSrc
2025-10-08 04:19:48,595 INFO lifecycle.LifecycleSupervisor: Stopping component: EventDrivenSourceRunner: { source:org.apache.flume.source.NetcatSource{name:netcatSrc,state:START} }
2025-10-08 04:19:48,595 INFO source.NetcatSource: Source stopping
2025-10-08 04:19:48,597 INFO node.Application: Stopping sink log
2025-10-08 04:19:48,597 INFO lifecycle.LifecycleSupervisor: Stopping component: SinkRunner: { policy:org.apache.flume.sink.DefaultSinkProcessor@7801c526 counterGroup:{ name:null counters:{runner.backoffs.consecutive=23, runner.backoffs=47} }}
2025-10-08 04:19:48,598 INFO node.Application: Stopping channel memChannel
2025-10-08 04:19:48,598 INFO lifecycle.LifecycleSupervisor: Stopping component: org.apache.flume.channel.MemoryChannel{name: memChannel}
2025-10-08 04:19:48,598 INFO instrumentation.MonitoredCounterGroup: Component type: CHANNEL, name: memChannel stopped
2025-10-08 04:19:48,598 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.start.time == 1759911233597
2025-10-08 04:19:48,598 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.stop.time == 1759911588598
2025-10-08 04:19:48,598 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.capacity == 100
2025-10-08 04:19:48,598 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.current.size == 0
2025-10-08 04:19:48,598 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.event.put.attempt == 3
2025-10-08 04:19:48,598 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.event.put.success == 3
2025-10-08 04:19:48,598 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.event.take.attempt == 50
2025-10-08 04:19:48,598 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.event.take.success == 3
2025-10-08 04:19:48,598 INFO lifecycle.LifecycleSupervisor: Stopping lifecycle supervisor 11
2025-10-08 04:19:48,634 INFO node.PollingPropertiesFileConfigurationProvider: configuration provider stopping
[root@osboxes flume]#

```

Рисунок 6 — Завершение работы агента

Таким образом, был произведен перенос данных через порт.

1.2 Перенос данных через буферную директорию

Создание конфигурации для agent2 представлено на Рисунке 7.

```

[root@osboxes flume]# vi transfer_spool.conf
[root@osboxes flume]# cat transfer_spool.conf
agent2.sources = dirSrc
agent2.channels = memChannel
agent2.sinks = filesink
agent2.sources.dirSrc.channels = memChannel
agent2.sinks.filesink.channel = memChannel
agent2.sources.dirSrc.type = spoolDir
agent2.sources.dirSrc.spoolDir = ./incoming
agent2.sinks.filesink.type = file_roll
agent2.sinks.filesink.sink.directory = ./output
agent2.sinks.filesink.sink.rollInterval = 0
agent2.channels.memChannel.type = memory
agent2.channels.memChannel.capacity = 100

```

Рисунок 7 — Конфигурация agent2

Процесс создания директорий представлен на Рисунке 8.

```

agent2.channels.memChannel.capacity = 100
[root@osboxes flume]# mkdir -p ~/flume/incoming ~/flume/output
[root@osboxes flume]#

```

Рисунок 8 — Процесс создания директорий

Запуск второго агента представлен на Рисунке 9.

```
root@osboxes: flume# flume-ng agent -name agent2 --conf-file transfer_spool.conf
Warning: No configuration directory set! Use --conf <dir> to override.
Info: Including Hadoop libraries found via (/home/hadoop/hadoop/bin/hadoop) for HDFS access
Info: Including HBASE libraries found via (/usr/local/hbase/hbase-2.3.5/bin/hbase) for HBASE access
Info: Including Hive libraries found via () for Hive access
+ exec /usr/lib/jvm/java-1.8.0-openjdk/bin/java -Xmx20m -cp '/usr/local/flume/flume-1.9.0/lib/*:/home/hadoop/hadoop/etc/hadoop:/home/hadoop/hadoop/share/hadoop/conf/*:/home/hadoop/hadoop/share/hadoop/hdfs/*:/home/hadoop/hadoop/share/hadoop/hdfs/lib/*:/home/hadoop/hadoop/share/hadoop/hdfs/lib/*:/home/hadoop/hadoop/share/hadoop/yarn/*:/home/hadoop/hadoop/share/hadoop/yarn/lib/*:/home/hadoop/hadoop-share/hadoop/yarn/*:/usr/local/hbase/hbase-2.3.5/conf/*:/usr/local/hbase/hbase-2.3.5:/usr/local/hbase/hbase-2.3.5/usr/local/hbase/hbase-2.3.5/lib/shaded-clients/hbase-shaded-client-byo-hadoop-2.3.5.jar:/usr/local/hbase/hbase-2.3.5/lib/client-cs-annotations-0.5.0.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/commons-logging-1.2.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/bat-jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/log4j-1.2.17.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/slf4j-api-1.7.30.jar:/home/hadoop/hadoop/hadoop/common/lib/*:/home/hadoop/hadoop/share/hadoop/common/*:/home/hadoop/hadoop/share/hadoop/hdfs/*:/home/hadoop/hadoop/share/hadoop/hadoop/hdfs/*:/home/hadoop/hadoop/share/hadoop/hdfs/*:/home/hadoop/hadoop/share/hadoop/mapreduce/*:/home/hadoop/hadoop/share/hadoop/yarn/*:/home/hadoop/hadoop/share/hadoop/yarn/lib/*:/home/hadoop/p/yam/*:/usr/local/hbase/hbase-2.3.5/conf/*:/lib/*' -Djava.library.path=/home/hadoop/hadoop/lib/native:/home/hadoop/hadoop/lib/native org.apache.flume.node.ApplicationTransferSpoolConf
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/flume/flume-1.9.0/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hadoop/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.30.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2025-10-08 04:29:33,953 INFO node.PollingPropertiesFileConfigurationProvider: Configuration provider starting
2025-10-08 04:29:33,961 INFO node.PollingPropertiesFileConfigurationProvider: Reloading configuration file:transfer_spool.conf
2025-10-08 04:29:33,960 INFO conf.FlumeConfiguration: Processing:dirSrc
2025-10-08 04:29:33,961 INFO conf.FlumeConfiguration: Processing:dirSrc
2025-10-08 04:29:33,961 INFO conf.FlumeConfiguration: Processing:FileSink
2025-10-08 04:29:33,961 INFO conf.FlumeConfiguration: Added sinks: fileSink Agent: agent2
2025-10-08 04:29:33,961 INFO conf.FlumeConfiguration: Processing:FileSink
2025-10-08 04:29:33,961 INFO conf.FlumeConfiguration: Processing:dirSrc
2025-10-08 04:29:33,961 INFO conf.FlumeConfiguration: Processing:memChannel
2025-10-08 04:29:33,962 INFO conf.FlumeConfiguration: Processing:FileSink
2025-10-08 04:29:33,962 INFO conf.FlumeConfiguration: Processing:memChannel
2025-10-08 04:29:33,962 INFO conf.FlumeConfiguration: Processing:FileSink
2025-10-08 04:29:33,962 WARN conf.FlumeConfiguration: Agent configuration for 'agent2' has no configfilters.
2025-10-08 04:29:33,985 INFO conf.FlumeConfiguration: Post-validation flume configuration contains configuration for agents: [agent2]
2025-10-08 04:29:33,986 INFO node.AbstractConfigurationProvider: Creating channels
2025-10-08 04:29:33,986 INFO channel.DefaultChannelFactory: Creating instance of channel memChannel type memory
2025-10-08 04:29:34,003 INFO node.AbstractConfigurationProvider: Created channel memChannel
2025-10-08 04:29:34,004 INFO source.DefaultSourceFactory: Creating instance of source dirSrc, type SpoolDir
2025-10-08 04:29:34,014 INFO sink.DefaultSinkFactory: Creating instance of sink: fileSink, type: file.roll
2025-10-08 04:29:34,020 INFO node.AbstractConfigurationProvider: Channel memChannel connected to [dirSrc,fileSink]
2025-10-08 04:29:34,026 INFO node.Application: Starting new configuration: { sourceRunners:[{dirSrc=EventDrivenSourceRunner: { source:Spool Directory source dirSrc:
}} sinkRunners:[{fileSink=SinkRunner: { policy:org.apache.flume.sink.DefaultSinkProcessor@8868abc counterGroup: { name:null counters:{}} }} channels:[{memChannel=or
moryChannel(name= memChannel)}]}
2025-10-08 04:29:34,026 INFO node.Application: Starting Channel memChannel
2025-10-08 04:29:34,236 INFO instrumentation.MonitoredCounterGroup: Monitored counter group for type: CHANNEL, name: memChannel: Successfully registered new MBean.
2025-10-08 04:29:34,236 INFO instrumentation.MonitoredCounterGroup: Component type: CHANNEL, name: memChannel started
2025-10-08 04:29:34,237 INFO node.Application: Starting Sink fileSink
2025-10-08 04:29:34,243 INFO sink.RollingFileSink: Starting org.apache.flume.sink.RollingFileSink{name=fileSink, channel=memChannel}...
2025-10-08 04:29:34,243 INFO node.Application: Starting Source dirSrc
2025-10-08 04:29:34,243 INFO instrumentation.MonitoredCounterGroup: Monitored counter group for type: SINK, name: fileSink: Successfully registered new MBean.
2025-10-08 04:29:34,243 INFO instrumentation.MonitoredCounterGroup: Component type: SINK, name: fileSink started
2025-10-08 04:29:34,243 INFO sink.RollingFileSink: RollInterval is not valid, file rolling will not happen.
2025-10-08 04:29:34,243 INFO sink.RollingFileSink: RollingFileSink fileSink started.
2025-10-08 04:29:34,244 INFO source.SpooledDirectorySource: SpooledDirectorySource source starting with directory: ./incoming
2025-10-08 04:29:34,282 INFO instrumentation.MonitoredCounterGroup: Monitored counter group for type: SOURCE, name: dirSrc: Successfully registered new MBean.
```

Рисунок 9 — Запуск agent2

Добавление файлов в директорию `incoming` представлено на Рисунке 10.

```
[root@osboxes incoming]# cd ~/flume/incoming
[root@osboxes incoming]# cp ~/data/*.txt .
[root@osboxes incoming]# vi hello.txt
[root@osboxes incoming]# cat hello.txt
```

Рисунок 10 — Добавление файлов в incoming

Проверка вывода агента представлена на Рисунке 11.

```
2025-10-08 04:29:34,282 INFO instrumentation.MonitoredCounterGroup: Component type: SOURCE, name: dirSrc started
2025-10-08 04:35:43,835 INFO avro.ReliableSpoolingFileEventReader: Last read took us just up to a file boundary. Rolling to the next file, if there is one.
2025-10-08 04:36:03,923 INFO avro.ReliableSpoolingFileEventReader: Preparing to move file /root/flume/.incoming/sample.txt to /root/flume/.incoming/sample.txt.COMPLETED
2025-10-08 04:36:03,923 INFO avro.ReliableSpoolingFileEventReader: Last read took us just up to a file boundary. Rolling to the next file, if there is one.
2025-10-08 04:36:03,923 INFO avro.ReliableSpoolingFileEventReader: Preparing to move file /root/flume/.incoming/hello.txt to /root/flume/.incoming/hello.txt.COMPLETED
```

Рисунок 11 — Проверка вывода агента

Проверка содержимого папки output представлено на Рисунке 12.

```
[root@osboxes flume]# cd output/
[root@osboxes output]# ls
1759912174017-1
[root@osboxes output]# cat 1759912174017-1
test
This is test file for Flume.
[root@osboxes output]# |
```

Рисунок 12 — Содержимое директории output

Завершение работы агента представлено на Рисунке 13.

```
2025-10-08 04:41:34,351 INFO node.Application: Stopping Sink fileSink
2025-10-08 04:41:34,351 INFO lifecycle.LifecycleSupervisor: Stopping component: SinkRunner: { policy:org.apache.flume.sink.DefaultSinkProcessor@78868abc coun
runner.backoffs.consecutive=91, runner.backoffs=91} }
2025-10-08 04:41:34,351 INFO sink.RollingFileSink: RollingFile sink fileSink stopping...
2025-10-08 04:41:34,351 INFO instrumentation.MonitoredCounterGroup: Component type: SINK, name: fileSink stopped
2025-10-08 04:41:34,351 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.start.time == 1759912174243
2025-10-08 04:41:34,351 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.stop.time == 1759912894351
2025-10-08 04:41:34,351 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.batch.complete == 0
2025-10-08 04:41:34,351 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.batch.empty == 0
2025-10-08 04:41:34,351 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.batch.underflow == 0
2025-10-08 04:41:34,351 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.channel.read.fail == 0
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.connection.closed.count == 0
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.connection.creation.count == 1
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.connection.failed.count == 0
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.event.drain.attempt == 2
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.event.drain.success == 2
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: SINK, name: fileSink. sink.event.write.fail == 0
2025-10-08 04:41:34,352 INFO sink.RollingFileSink: RollingFile sink fileSink stopped. Event metrics: SINK:fileSink[sink.batch.underflow=0, sink.event.write.f
ount=0, sink.channel.read.fail=0, sink.connection.closed.count=1, sink.event.drain.attempt=2, sink.batch.complete=0, sink.event.drain.success=2, sink.connecti
.empty=0]
2025-10-08 04:41:34,352 INFO node.Application: Stopping Channel memChannel
2025-10-08 04:41:34,352 INFO lifecycle.LifecycleSupervisor: Stopping component: org.apache.flume.channel.MemoryChannel{name: memChannel}
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Component type: CHANNEL, name: memChannel stopped
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.start.time == 1759912174236
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.stop.time == 1759912894352
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.capacity == 100
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.current.size == 0
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.event.put.attempt == 2
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.event.put.success == 2
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.event.take.attempt == 94
2025-10-08 04:41:34,352 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel. channel.event.take.success == 2
2025-10-08 04:41:34,352 INFO lifecycle.LifecycleSupervisor: Stopping lifecycle supervisor 11
2025-10-08 04:41:34,356 INFO node.PollingPropertiesFileConfigurationProvider: Configuration provider stopping
[root@osboxes flume]#
```

Рисунок 13 — Завершение работы агента

Таким образом, были перенесены данные через буферную директорию

1.3 Использование перехватчика

Создание конфигурации перехватчика представлено на Рисунке 14.

```
[root@osboxes ~]# cd flume
[root@osboxes flume]# ls
incoming output transfer.conf transfer_spool.conf
[root@osboxes flume]# vi interceptor.conf
[root@osboxes flume]# cat interceptor.conf
agent3.sources = netcatSrc
agent3.channels = memChannel
agent3.sinks = log
agent3.sources.netcatSrc.channels = memChannel
agent3.sinks.log.channel = memChannel
agent3.sources.netcatSrc.type = netcat
agent3.sources.netcatSrc.bind = 0.0.0.0
agent3.sources.netcatSrc.port = 3333
agent3.sinks.log.type = logger
agent3.channels.memChannel.type = memory
agent3.channels.memChannel.capacity = 100
agent3.sources.netcatSrc.interceptors = i1
agent3.sources.netcatSrc.interceptors.i1.type = host
agent3.sources.netcatSrc.interceptors.i1.hostHeader = hostname
[root@osboxes flume]# |
```

Рисунок 14 — Конфигурация перехватчика

Запуск agent3 представлен на Рисунке 15.

```
[root@osboxes flume]# flume-ng agent -name agent3 --conf-file interceptor.conf
Warning: No configuration directory set! Use -conf <dir> to override.
Info: Including Hadoop libraries found via (/home/hadoop/hadoop/bin/hadoop) for HDFS access
Info: Including HBase libraries found via (/usr/local/hbase/hbase-2.3.5/bin/hbase) for HBASE access
Info: Including Hive libraries found via O for Hive access
+ exec /usr/lib/jvm/java-1.8.0-openjdk/bin/java -Xmx20m -cp '/usr/local/flume/flume-1.9.0/lib/*:/home/hadoop/hadoop/etc/hadoop:/home/hadoop/hadoop/share/hadoop/common/share/hadoop/common/*:/home/hadoop/hadoop/share/hadoop/hdfs:/home/hadoop/hadoop/share/hadoop/hdfs/lib/*:/home/hadoop/hadoop/share/hadoop/hdfs/*:/home/hadoop/hadoop/share/hadoop/yarn/*:/usr/local/hbase/hbase-2.3.5/conf:/usr/local/hbase/hbase-2.3.5/lib/shaded-clients/hbase-shaded-client-byo-hadoop-2.3.5.jar:/usr/local/hbase/hbase-2.3.5/lib/client-ce-annotations-0.5.0.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/commons-logging-1.2.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/batting.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/log4j-1.2.17.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/slf4j-api-1.7.30.jar:/home/hadoop/hadoop/share/hadoop/common/lib/*:/home/hadoop/hadoop/share/hadoop/yarn/lib/*:/home/hadoop/hadoop/share/hadoop/hdfs:/home/hadoop/hadoop/share/hadoop/hdfs/lib/*:/home/hadoop/hadoop/share/hadoop/hdfs/*:/home/hadoop/hadoop/share/hadoop/mapreduce/*:/home/hadoop/hadoop/share/hadoop/yarn:/home/hadoop/hadoop/share/hadoop/yarn/lib/*:/home/hadoop/hadoop/share/hadoop/yarn/*:/usr/local/hbase/hbase-2.3.5/conf:/lib/*' -Djava.library.path=/home/hadoop/hadoop/lib/native:/home/hadoop/hadoop/lib/native org.apache.flume.node.ApplicationInterceptor
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/flume/flume-1.9.0/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/home/hadoop/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.30.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2025-10-08 06:57:04,521 INFO node.PollingPropertiesFileConfigurationProvider: Configuration provider starting
2025-10-08 06:57:04,526 INFO node.PollingPropertiesFileConfigurationProvider: Reloading configuration file:interceptor.conf
2025-10-08 06:57:04,529 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 06:57:04,529 INFO conf.FlumeConfiguration: Processing:log
2025-10-08 06:57:04,529 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 06:57:04,529 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 06:57:04,530 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 06:57:04,530 INFO conf.FlumeConfiguration: Processing:log
2025-10-08 06:57:04,530 INFO conf.FlumeConfiguration: Processing:memChannel
2025-10-08 06:57:04,530 INFO conf.FlumeConfiguration: Processing:memChannel
2025-10-08 06:57:04,530 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 06:57:04,530 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 06:57:04,530 INFO conf.FlumeConfiguration: Added sinks: log Agent: agent3
2025-10-08 06:57:04,530 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 06:57:04,530 WARN conf.FlumeConfiguration: Agent configuration for 'agent3' has no configfilters.
2025-10-08 06:57:04,531 INFO conf.FlumeConfiguration: Post-validation flume configuration contains configuration for agents: [agent3]
2025-10-08 06:57:04,551 INFO node.AbstractConfigurationProvider: Creating channels
2025-10-08 06:57:04,557 INFO channel.DefaultChannelFactory: Creating instance of channel memChannel type memory
2025-10-08 06:57:04,561 INFO node.AbstractConfigurationProvider: Created channel memChannel
2025-10-08 06:57:04,562 INFO source.DefaultSourceFactory: Creating instance of source netcatSrc, type netcat
2025-10-08 06:57:04,588 INFO sink.DefaultSinkFactory: Creating instance of sink: log, type: logger
2025-10-08 06:57:04,592 INFO node.AbstractConfigurationProvider: channel memChannel connected to [netcatSrc, log]
2025-10-08 06:57:04,599 INFO node.Application: Starting new configuration: { sourceRunners:{netcatSrc=EventDrivenSourceRunner: { source:org.apache.flume.source.NetcatSource, state:START } } sinkRunners:{log=SinkRunner: { policy:org.apache.flume.sink.DefaultSinkProcessor@444c70e3 counterGroup: { name:null counters:{ } } } } channels:{memChannel=MemoryChannel{name: memChannel}}}
2025-10-08 06:57:04,600 INFO node.Application: Starting Channel memChannel
2025-10-08 06:57:04,802 INFO instrumentation.MonitoredCounterGroup: Monitored counter group for type: CHANNEL, name: memChannel: Successfully registered new MBean.
2025-10-08 06:57:04,802 INFO instrumentation.MonitoredCounterGroup: Component type: CHANNEL, name: memChannel started
2025-10-08 06:57:04,803 INFO node.Application: Starting Sink log
2025-10-08 06:57:04,804 INFO node.Application: Starting Source netcatSrc
2025-10-08 06:57:04,806 INFO source.NetcatSource: Source starting
2025-10-08 06:57:04,819 INFO source.NetcatSource: Created serverSocket:sun.nio.ch.ServerSocketChannelImpl[/0:0:0:0:0:0:0:0:3333]
```

Рисунок 15 — Запуск перехватчика

Принятие данных перехватчиком представлено на Рисунке 16.

```
2025-10-08 06:57:58,863 INFO sink.LoggerSink: Event: { headers:{hostname=127.0.0.1} body: 54 68 69 73 20 69 73 20 74 65 73 74 69 6E 67 20 This is testing }
2025-10-08 06:57:59,064 INFO sink.LoggerSink: Event: { headers:{hostname=127.0.0.1} body: 48 61 64 6F 6F 70 0D Hadoop. }
```

Рисунок 16 — Получение данных с выводом ip отправителя

Очистка временных данных представлена на Рисунке 17.

```
2025-10-08 06:59:45,780 INFO node.Application: Shutting down configuration: { sourceRunners:{netcatSrc=EventDrivenSourceRunner: { source:org.apache.flume.source.NetcatSource, state:START } } sinkRunners:{log=SinkRunner: { policy:org.apache.flume.sink.DefaultSinkProcessor@444c70e3 counterGroup: { name:null counters:{ } } } } channels:{memChannel=org.apache.flume.channel.MemoryChannel{name: memChannel}}}
2025-10-08 06:59:45,780 INFO node.Application: Stopping Source netcatSrc
2025-10-08 06:59:45,780 INFO lifecycle.LifecycleSupervisor: Stopping component: EventDrivenSourceRunner: { source:org.apache.flume.source.NetcatSource, state:START }
2025-10-08 06:59:45,780 INFO source.NetcatSource: Source stopping
2025-10-08 06:59:45,781 INFO node.Application: Stopping Sink log
2025-10-08 06:59:45,781 INFO lifecycle.LifecycleSupervisor: Stopping component: SinkRunner: { policy:org.apache.flume.sink.DefaultSinkProcessor@444c70e3 counterGroup: { name:null counters:{ } } }
2025-10-08 06:59:45,783 INFO node.Application: Stopping Channel memChannel
2025-10-08 06:59:45,783 INFO lifecycle.LifecycleSupervisor: Stopping component: org.apache.flume.channel.MemoryChannel
2025-10-08 06:59:45,783 INFO instrumentation.MonitoredCounterGroup: Component type: CHANNEL, name: memChannel stopped
2025-10-08 06:59:45,783 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel
2025-10-08 06:59:45,784 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel
2025-10-08 06:59:45,784 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel
2025-10-08 06:59:45,784 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel
2025-10-08 06:59:45,784 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel
2025-10-08 06:59:45,784 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel
2025-10-08 06:59:45,784 INFO instrumentation.MonitoredCounterGroup: Shutdown Metric for type: CHANNEL, name: memChannel
2025-10-08 06:59:45,784 INFO lifecycle.LifecycleSupervisor: Stopping lifecycle supervisor 11
2025-10-08 06:59:45,796 INFO node.PollingPropertiesFileConfigurationProvider: Configuration provider stopping
[root@osboxes flume]# cd ~/flume
[root@osboxes flume]# rm -rf incoming output
[root@osboxes flume]# |
```

Рисунок 17 — Очистка временных данных

Таким образом, была произведена работа с перехватчиком.

2 САМОСТОЯТЕЛЬНАЯ РАБОТА

Конфигурация для agent4 представлена на Рисунке 18.

```
[root@osboxes flume]# vi custom.conf
[root@osboxes flume]# cat custom.conf
agent4.sources = netcatSrc
agent4.channels = memChannel
agent4.sinks = log
agent4.sources.netcatSrc.channels = memChannel
agent4.sinks.log.channel = memChannel
agent4.sources.netcatSrc.type = netcat
agent4.sources.netcatSrc.bind = 0.0.0.0
agent4.sources.netcatSrc.port = 11111
agent4.sinks.log.type = logger
agent4.channels.memChannel.type = file
agent4.channels.memChannel.capacity = 1000
agent4.channels.memChannel.transactionCapacity = 100
[root@osboxes flume]# |
```

Рисунок 18 — Конфигурация для agent4

Запуск агента представлен на Рисунке 19.

```

agent4.channels.memChannel.transactionCapacity = 1000
[root@osboxes flume]# flume-ng agent -name agent4 --conf-file custom.conf
Warning: No configuration directory set! Use --conf <dir> to override.
Info: Including Hadoop libraries found via (/home/hadoop/hadoop/bin/hadoop) for HDFS access
Info: Including HBASE libraries found via (/usr/local/hbase/hbase-2.3.5/bin/hbase) for HBASE access
Info: Including Hive libraries found via () for Hive access
+ exec /usr/lib/jvm/java-1.8.0-openjdk/bin/java -Xmx20m -cp '/usr/local/flume/flume-1.9.0/lib/*:/home/hadoop/hadoop/etc/hadoop:/home
share/hadoop/hdfs:/home/hadoop/hadoop/share/hadoop/hdfs/lib/*:/home/hadoop/hadoop/share/hadoop/hdfs/*:/home/hadoop/hadoop/share/hado
hadoop/hadoop/share/hadoop/yarn/*:/usr/local/hbase/hbase-2.3.5/conf:/usr/lib/jvm/jre/lib/tools.jar:/usr/local/hbase/hbase-2.3.5/us
hbase-2.3.5/lib/client-facing-thirdparty/audience-annotations-0.5.0.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/co
bating.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing-thirdparty/log4j-1.2.17.jar:/usr/local/hbase/hbase-2.3.5/lib/client-facing
/lib/*:/home/hadoop/hadoop/share/hadoop/common/*:/home/hadoop/hadoop/share/hadoop/hdfs:/home/hadoop/hadoop/share/hadoop/hdfs/lib/*:/
/share/hadoop/yarn:/home/hadoop/hadoop/share/hadoop/yarn/lib/*:/home/hadoop/hadoop/share/hadoop/yarn/*:/usr/local/hbase/hbase-2.3.5/
che.flume.node.Application -name agent4 --conf-file custom.conf
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/flume/flume-1.9.0/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class
SLF4J: Found binding in [jar:file:/home/hadoop/hadoop/share/hadoop/common/lib/slf4j-log4j12-1.7.30.jar!/org/slf4j/impl/StaticLoggerB
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
2025-10-08 07:31:14,878 INFO node.PollingPropertiesFileConfigurationProvider: Configuration provider starting
2025-10-08 07:31:14,882 INFO node.PollingPropertiesFileConfigurationProvider: Reloading configuration file:custom.conf
2025-10-08 07:31:14,885 INFO conf.FlumeConfiguration: Processing:memChannel
2025-10-08 07:31:14,885 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 07:31:14,885 INFO conf.FlumeConfiguration: Added sinks: log Agent: agent4
2025-10-08 07:31:14,885 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 07:31:14,885 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 07:31:14,885 INFO conf.FlumeConfiguration: Processing:memChannel
2025-10-08 07:31:14,885 INFO conf.FlumeConfiguration: Processing:memChannel
2025-10-08 07:31:14,886 INFO conf.FlumeConfiguration: Processing:log
2025-10-08 07:31:14,886 INFO conf.FlumeConfiguration: Processing:log
2025-10-08 07:31:14,887 INFO conf.FlumeConfiguration: Processing:netcatSrc
2025-10-08 07:31:14,887 WARN conf.FlumeConfiguration: Agent configuration for 'agent4' has no configfilters.
2025-10-08 07:31:14,908 INFO conf.FlumeConfiguration: Post-validation flume configuration contains configuration for agents: [agent4
2025-10-08 07:31:14,908 INFO node.AbstractConfigurationProvider: Creating channels
2025-10-08 07:31:14,914 INFO channel.DefaultChannelFactory: Creating instance of channel memChannel type file
2025-10-08 07:31:14,932 INFO node.AbstractConfigurationProvider: Created channel memChannel
2025-10-08 07:31:14,932 INFO source.DefaultSourceFactory: Creating instance of source netcatSrc, type netcat
2025-10-08 07:31:14,938 INFO sink.DefaultSinkFactory: Creating instance of sink: log, type: logger
2025-10-08 07:31:14,942 INFO node.AbstractConfigurationProvider: channel memChannel connected to [netcatSrc, log]
2025-10-08 07:31:14,947 INFO node.Application: Starting new configuration:{ sourceRunners:{netcatSrc=EventDrivenSourceRunner: { sour
lcy.org.apache.flume.sink.DefaultSinkProcessor@59bea875 counterGroup:{ name:null counters:{} } }} channels:{memChannel=FileChannel
2025-10-08 07:31:14,951 INFO node.Application: Starting channel memChannel
2025-10-08 07:31:14,952 INFO file.FileChannel: Starting FileChannel memChannel { dataDirs: [/root/.flume/file-channel/data] }...
2025-10-08 07:31:15,105 INFO instrumentation.MonitoredCounterGroup: Monitored counter group for type: CHANNEL, name: memChannel: suc
2025-10-08 07:31:15,105 INFO instrumentation.MonitoredCounterGroup: Component type: CHANNEL, name: memChannel started
2025-10-08 07:31:15,119 INFO file.Log: Encryption is not enabled
2025-10-08 07:31:15,120 INFO file.Log: Replay started
2025-10-08 07:31:15,121 INFO file.Log: Found NextFileID 0, from []
2025-10-08 07:31:15,131 INFO file.EventQueueBackingStoreFile: Preallocated /root/.flume/file-channel/checkpoint/checkpoint to 16232
2025-10-08 07:31:15,132 INFO file.EventQueueBackingStoreFileV3: Starting up with /root/.flume/file-channel/checkpoint/checkpoint and
2025-10-08 07:31:15,212 INFO file.FlumeEventQueue: QueueSet population inserting 0 took 0
2025-10-08 07:31:15,215 INFO file.Log: Last checkpoint wed Oct 08 07:31:15 EDT 2025, queue depth = 0
2025-10-08 07:31:15,216 INFO file.Log: Replaying logs with v2 replay logic
2025-10-08 07:31:15,217 INFO file.ReplayHandler: Starting replay of []
2025-10-08 07:31:15,218 INFO file.ReplayHandler: read: 0, put: 0, take: 0, rollback: 0, commit: 0, skip: 0, eventCount:0
2025-10-08 07:31:15,218 INFO file.FlumeEventQueue: Search Count = 0, Search Time = 0, Copy Count = 0, Copy Time = 0
2025-10-08 07:31:15,229 INFO file.Log: Rolling /root/.flume/file-channel/data
2025-10-08 07:31:15,229 INFO file.Log: Roll start /root/.flume/file-channel/data
2025-10-08 07:31:15,235 INFO tools.DirectMemoryUtils: Unable to get maxDirectMemory from VM: NoSuchMethodException: sun.misc.VM.maxD
2025-10-08 07:31:15,236 INFO tools.DirectMemoryUtils: Direct Memory Allocation: Allocation = 1048576, Allocated = 0, MaxDirectMemor
2025-10-08 07:31:15,286 INFO file.LogFile: Opened /root/.flume/file-channel/data/log-1
2025-10-08 07:31:15,296 INFO file.Log: Roll end
2025-10-08 07:31:15,296 INFO file.EventQueueBackingStoreFile: Start checkpoint for /root/.flume/file-channel/checkpoint/checkpoint,
2025-10-08 07:31:15,305 INFO file.EventQueueBackingStoreFile: Updating checkpoint metadata: logwriteorderID: 1759923075230, queueSiz
2025-10-08 07:31:15,328 INFO file.Log: Updated checkpoint for file: /root/.flume/file-channel/data/log-1 position: 0 logwriteorderID
2025-10-08 07:31:15,328 INFO file.FileChannel: Queue Size after replay: 0 [channel=memChannel]
2025-10-08 07:31:15,329 INFO node.Application: Starting sink log
2025-10-08 07:31:15,330 INFO node.Application: Starting source netcatSrc
2025-10-08 07:31:15,331 INFO source.NetcatSource: Source starting
2025-10-08 07:31:15,342 INFO source.NetcatSource: Created serverSocket:sun.nio.ch.ServerSocketChannelImpl[/0:0:0:0:0:0:0:11111]

```

Рисунок 19 — Запуск агента agent4

Проверка отправки сообщения представлена на Рисунке 20.

```

2025-10-08 07:31:15,342 INFO source.NetcatSource: Created serverSocket:sun.nio.ch.ServerSocketChannelImpl[/0:0:0:0:0:0:0:11111]
2025-10-08 07:32:50,417 INFO sink.LoggerSink: Event: { headers:{} body: 48 65 6C 6C 6F 20 77 6F 72 6C 64 21 0D Hello world!. }

```

Рисунок 20 — Проверка принятия сообщения

Конфигурация для перехватчика у агента agent 4 представлена на Рисунке

21.

```

[root@osboxes flume]# cat custom.conf
agent4.sources = netcatSrc
agent4.channels = memChannel
agent4.sinks = log
agent4.sources.netcatSrc.channels = memChannel
agent4.sinks.log.channel = memChannel
agent4.sources.netcatSrc.type = netcat
agent4.sources.netcatSrc.bind = 0.0.0.0
agent4.sources.netcatSrc.port = 11111
agent4.sinks.log.type = logger
agent4.channels.memChannel.type = file
agent4.channels.memChannel.capacity = 1000
agent4.channels.memChannel.transactionCapacity = 100
agent4.sources.netcatSrc.interceptors = i1
agent4.sources.netcatSrc.interceptors.i1.type = org.apache.flume.interceptor.TimestampInterceptor$Builder
agent4.sinks.log.filePrefix = FlumeData.%{CollectorHost}.%Y-%m-%d
[root@osboxes flume]#

```

Рисунок 21 — Перехватчик для agent4

Вывод с timestamp представлен на Рисунке 22.

```
2025-10-08 07:36:28,740 INFO sink.LoggerSink: Event: { headers:{timestamp=1759923385183} body: 48 65 6C 6C 6F 20 57 6F 72 6C 64 21 0D Hello world!. }
```

Рисунок 22 — Вывод сообщения с перехватчиком

Таким образом был настроен агент с добавлением перехватчика.

ВЫВОД

В ходе выполнения практической работы №3 по дисциплине «Системы управления большими данными» были освоены основные принципы работы с Apache Flume — распределённой, надёжной и доступной системой для эффективного сбора, агрегации и перемещения больших объёмов лог-данных. В рамках лабораторной работы последовательно реализованы три базовых сценария: передача данных через сетевой порт с использованием источника netcat, перенос файлов через буферную директорию с применением источника spooldir и использование перехватчика (interceptor) для модификации событий — в частности, добавления метки времени и информации об IP-адресе отправителя. Кроме того, в разделе самостоятельной работы был разработан и протестирован собственный конфигурационный файл агента, включающий настройку перехватчика для автоматического добавления временной метки к каждому событию. Все этапы сопровождались визуальной проверкой корректности работы агентов, анализом вывода и содержимого целевых директорий. В результате выполнения работы подтверждена работоспособность различных компонентов Flume, а также продемонстрирована гибкость системы в адаптации под различные сценарии сбора и предварительной обработки потоковых данных.